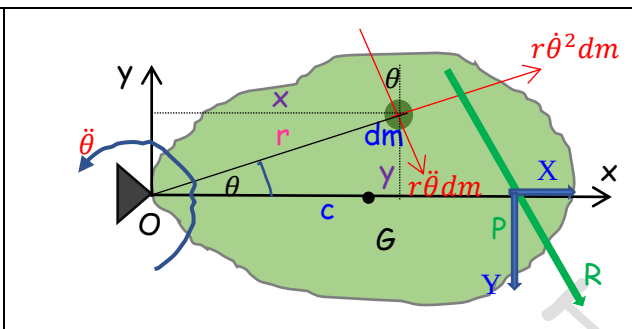


7.4 Resultant Inertia Force in Rotation

Consider a rigid body rotating counter clockwise about an axis perpendicular to the x-y plane and passing through O. Let the centre of gravity G of the body be at a distance c from the axis of rotation. Choose the co-ordinate system in such a way that the x-axis passes through the centre of gravity G.



Consider an elemental mass dm at a distance r from the axis of rotation. Let the body has an angular velocity $\dot{\theta}$ and an angular acceleration $\ddot{\theta}$. The mass element will be having a normal acceleration $r\dot{\theta}^2$ directed towards the centre of rotation and a tangential acceleration $r\ddot{\theta}$ tangential to the circular path. By D'Alembert's principle, the mass element will be experiencing a normal inertia force $r\dot{\theta}^2 dm$ directed opposite to the direction of normal acceleration and a tangential inertia force $r\ddot{\theta} dm$ opposite to the direction of tangential acceleration.

The x-component of the elemental inertia forces, $dX = r\dot{\theta}^2 dm \cos \theta + r\ddot{\theta} dm \sin \theta$

The y-component of the elemental inertia forces, $dY = r\dot{\theta}^2 dm \sin \theta - r\ddot{\theta} dm \cos \theta$

But, $r \cos \theta = x$ and $r \sin \theta = y$

$$dX = \dot{\theta}^2 x dm + \ddot{\theta} y dm$$

$$dY = \dot{\theta}^2 y dm - \ddot{\theta} x dm$$

The x-component of the resultant inertia forces of infinite number of such mass elements,

$$X = \int dX = \int \dot{\theta}^2 x dm + \ddot{\theta} y dm = \dot{\theta}^2 \int x dm + \ddot{\theta} \int y dm = \dot{\theta}^2 \bar{x} \cdot m + \ddot{\theta} \bar{y} \cdot m$$

Where m is the mass of the body. $\bar{x}$ and $\bar{y}$ are the x and y co-ordinates of the centre of gravity G of the body. But, the x-axis is chosen to pass through the centre of gravity.

Therefore, $\bar{y} = 0$ and $\bar{x} = c$.

$$\therefore X = \dot{\theta}^2 \bar{x} \cdot m = \frac{W}{g} c \dot{\theta}^2$$

Similarly, the y-component of the resultant inertia forces

$$Y = \int dY = \int \dot{\theta}^2 y dm - \ddot{\theta} x dm = \dot{\theta}^2 \int y dm - \ddot{\theta} \int x dm = \dot{\theta}^2 \bar{y} \cdot m - \ddot{\theta} \bar{x} \cdot m$$

$$\therefore Y = -\ddot{\theta} \bar{x} \cdot m = -\frac{W}{g} c \ddot{\theta}$$

The negative sign shows that the Y component of the resultant inertia forces is in the negative y-direction.

The magnitude of the x-component of the resultant inertia force, $X = \frac{W}{g} c \dot{\theta}^2$.

This is same as the normal inertia force developed if the mass $\frac{W}{g}$ is concentrated at the centre of gravity.

The magnitude of the y-component of the resultant inertia force, $Y = \frac{W}{g} c \ddot{\theta}$

This is same as the tangential inertia force developed if the mass $\frac{W}{g}$ is concentrated at the centre of gravity. (This is possible only if the x-axis passes through the centre of gravity).

The resultant inertia force, $R = \sqrt{X^2 + Y^2}$.

It will be acting at an angle given by $\tan \theta = \frac{Y}{X}$

To find the line of action of this resultant inertia force, let us apply the Varignon's theorem of moments. Let this resultant R intersect the x -axis at P .

Moment of the resultant inertia force about O =

algebraic sum of the moments of the components X and Y about the same point.

$$= -Y \cdot OP$$

$$-Y \cdot OP = - \int (r \ddot{\theta} dm) \times r = -\ddot{\theta} \int r^2 dm = -I_O \ddot{\theta}$$

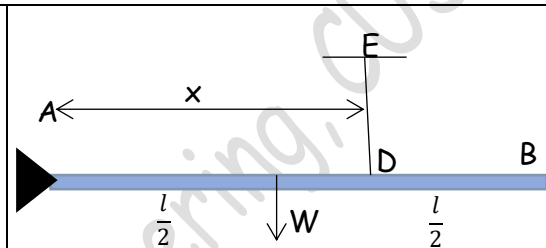
$$\text{i.e., } \frac{W}{g} c \ddot{\theta} \times OP = I_O \ddot{\theta} = \frac{W}{g} i_o^2 \ddot{\theta}$$

$$OP = \frac{i_o^2}{c} = c + \frac{i_g^2}{c}$$

i.e., the resultant inertia force passes through the centre of percussion.

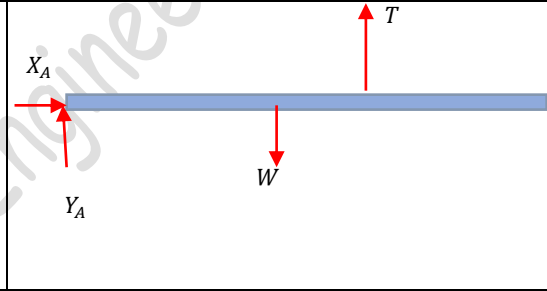
Example 1

A slender bar AB is supported in a horizontal position as shown. At what distance x from the hinge A should the vertical string DE be attached to the bar in order that when it is cut, there will be no immediate change in the reaction at A ?

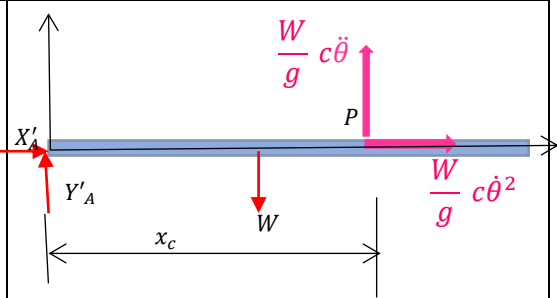


Consider the bar in static equilibrium. The free-body diagram of the bar is shown in the figure.

$$\begin{aligned} \Sigma F_x = 0 &\gg X_A = 0 \\ \Sigma F_y = 0 &\gg Y_A + T - W = 0 \\ \Sigma M_A = 0 &\gg T \times x - W \times \frac{l}{2} = 0 \gg T = \frac{Wl}{2x} \\ Y_A &= \frac{Wl}{2x} - W \end{aligned}$$



Now consider that the string is cut. The bar starts rotating downwards with respect to the axis through A . Choosing the x -axis passing through the centre of gravity and y -axis perpendicular to it. The resultant inertia force is resolved in to its components $\frac{W}{g} c \dot{\theta}^2$ and $\frac{W}{g} c \ddot{\theta}$ acting at the centre of percussion P .



Immediately after the string is cut, the angular velocity of the bar is zero. So, the resultant normal inertia force in the x -direction is zero. Since the inertia forces are incorporated in to the system, the system is in dynamic equilibrium. So we can write,

$$\Sigma M_A = 0 \gg \frac{W}{g} c \ddot{\theta} \times x_c - W \times \frac{l}{2} = 0$$

$$\Sigma F_x = 0 \gg X'_A = 0$$

$$\Sigma F_y = 0 \gg Y'_A + \frac{W}{g} c \ddot{\theta} - W = 0$$

$$c = \frac{l}{2} \quad \text{and} \quad x_c = \frac{i_A^2}{c}$$

Moment of inertia of the bar with respect to the axis through its end, $I_{AA} = \frac{W l^2}{g \cdot 3}$

$$I_{AA} = \frac{W}{g} i_A^2 = \frac{W l^2}{g \cdot 3} \gg i_A^2 = \frac{l^2}{3}$$

$$x_c = \frac{i_A^2}{c} = \frac{\frac{l^2}{3}}{\frac{l}{2}} = \frac{2}{3}l$$

$$\frac{W}{g} \frac{l}{2} \ddot{\theta} \times \frac{2}{3}l - W \times \frac{l}{2} = 0 \gg \gg \ddot{\theta} = \frac{3g}{2l}$$

$$Y'_A + \frac{W}{g} c \ddot{\theta} - W = 0 \gg \gg Y_A = W - \frac{W}{g} c \ddot{\theta} = W - \frac{W}{g} \frac{l}{2} \frac{3g}{2l}$$

$$Y'_A = \frac{W}{4}$$

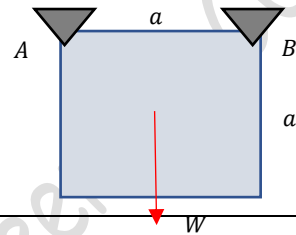
For there is to be no change in the reaction,

$$\frac{Wl}{2x} - W = \frac{W}{4}$$

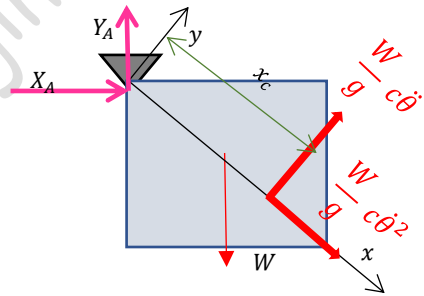
$$x = \frac{2}{3}l$$

Example 2

A homogeneous square plate of weight W and side a hangs in a vertical plane by two pins A and B as shown. Calculate the horizontal and vertical components of the reaction at A an instant immediately after the pin at B is removed.



When the pin at B is removed, the plate starts swinging down clockwise about the pin at A . The x -axis is drawn along the centre of gravity. The resultant normal inertia force $\frac{W}{g} c \ddot{\theta}^2$ and tangential inertia force $\frac{W}{g} c \ddot{\theta}$ are incorporated along the x and y axis from the centre of percussion.



Immediately after the pin at B is removed, the plate starts swinging from rest. Therefore, its initial angular velocity is zero. Hence, its resultant normal inertia force is equal to zero.

$$\Sigma M_A = 0 \gg \gg \frac{W}{g} c \ddot{\theta} \times x_c - W \times \frac{a}{2} = 0$$

$$\Sigma F_x = 0 \gg \gg X_A + \frac{W}{g} c \ddot{\theta} \cos 45 = 0$$

$$\Sigma F_y = 0 \gg \gg Y_A + \frac{W}{g} c \ddot{\theta} \sin 45 - W = 0$$

$$c = \frac{a}{\sqrt{2}}$$

The mass moment of inertia of a square plate with respect to the perpendicular axis passing through its centroid

$$I_{GG} = \frac{W}{g} \frac{a^2}{6} \gg \gg \frac{W}{g} i_g^2 = \frac{W}{g} \frac{a^2}{6}$$

$$i_g^2 = \frac{a^2}{6}$$

$$x_c = c + \frac{i_g^2}{c} = \frac{a}{\sqrt{2}} + \frac{a^2/6}{a/\sqrt{2}} = \frac{4a}{3\sqrt{2}}$$

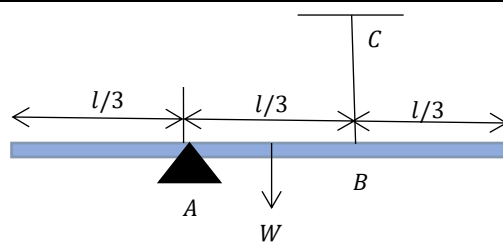
$$\frac{W}{g} \frac{a}{\sqrt{2}} \ddot{\theta} \times \frac{4a}{3\sqrt{2}} - W \times \frac{a}{2} = 0 \gg \gg \ddot{\theta} = \frac{3g}{4a}$$

$$X_A + \frac{W}{g} \frac{a}{\sqrt{2}} \frac{3g}{4a} \cos 45 = 0 \gg \gg X_A = -\frac{3W}{8}$$

$$Y_A + \frac{W}{g} \frac{a}{\sqrt{2}} \frac{3g}{4a} \sin 45 - W = 0 \gg \gg Y_A = \frac{5W}{8}$$

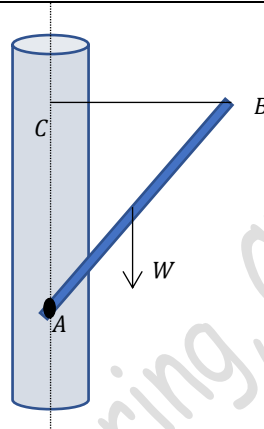
#Exercise 1

A slender prismatic bar of length l and weight W is supported in a horizontal position as shown in the figure. Find the instantaneous reaction at A immediately after the string BC is cut.



#Exercise 2

A slender bar AB of length $l = 1.2\text{m}$ and weight $W = 89\text{ N}$ is attached to a vertical shaft as shown. Assuming the system is at rest, calculate the instantaneous values for the components of the reaction at A just after the string BD is cut.



Prof. Biju N, School of Engineering, CUSAT